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Contract Checking as Effects

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ABSTRACT

Contract Checking as Effects

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Acknowledgements

Text for acknowledgments goes here. You can thank your advisors, peers, and family.

Preface

This is the preface (optional).

Dedication

This is the dedication (optional).

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CHAPTER 1

Introduction

Software contracts, or *behavioral contracts*, allow programmers to attach assertions to values, and functions. These assertions are runtime guarantees, on violation, an error is thrown, blaming the violating party. Contracts are most useful when placed on a function, enforcing the function’s input and output values. While first-order functional contracts are useful, they are too simple. Higher-order functional contracts allows programmers to enforce contracts for higher-order functions. Dependent contracts are functional contracts allowing usage of the function argument during post-condition contract checking.

Recently, algebraic effect handlers has gained a large attraction, and is adopted in OCaml since version 5.0. Effect handlers open a new way to identify effects and handling them. Handlers are defined to handle one or many effects. A handler for an effect can stop the program or continue with some result. Handlers do not have state by default, but it can be implemented by applying the new states to the continuation closure, and let the underlying computation receives a state argument.

Combining the contract system and effect handlers is a new approach. Effectful Contract [9] demonstrates how both systems can be combined, providing contracts for effectful code. Contracts in their system remain a language feature. In this dissertation, instead of combining the effects and contracts, we ask ourselves whether contract checking itself can become an effect, and reuse the capabilities of handlers.

Contracts checking can be encoded as side-effects and benefited from context sharing when handled using effect handlers.

1.1 Contributions

This thesis explores the contracts checking as effects’ approach to implementing software contracts. Through experiments, we show that this approach accomodates functional contracts, dependent contracts, and the challenging higher-order contracts. We also discovered that as effects, their contexts can be shared if they are under the same handler. Sharing contracts context opens new capabilities to optimizations and contract state sharing. This disseration shows the above idea by providing a compiler model between CPCF [2] and System F^ε [14]. We also provide an implementation in OCaml as a practical implementation.

The rest of the thesis is organized as follows: Chapter [ref{chap:contracts}](#) and Chapter [ref{chap:effects}](#) introduces the two formal languages [Contracts](#) based on CPCF, and [Effects](#) based on System F^ε ; Chapter [ref{chap:compiler}](#) discusses a theoretical compiler from [Contracts](#) to [Effects](#) as a demonstration to contract checking as effects’; Chapter [ref{chap:ocaml}](#) introduces the OCaml language and its native support for effect handlers; Chapter [ref{chap:implementation}](#) provides the implementation of the theoretical compiler in OCaml using PPX rewriter; Chapter [ref{chap:compare}](#) compares our implementation in OCaml to the Racket’s contract system; Chapter [ref{chap:final}](#) concludes our remarks.

CHAPTER 2

Formal Language Models

2.1 Base Language

We define Base language and its syntax in Figure 1. The language extends the PCF language with tuples and mutable cells. This will be the base language for both **Contracts** and **Effects**.

Base	
$\tau ::= o$ $\quad \tau \rightarrow \tau$ $\quad \langle \tau, \tau \rangle$ $\quad \text{ref}(\tau)$	$o ::= \text{num}$ $\quad \text{bool}$
$e ::= 0 \mid -1 \mid 1 \mid \dots \mid \text{tt} \mid \text{ff} \mid x$ $\quad \lambda x : \tau. e \mid \mu x : \tau. e \mid \text{loc}$ $\quad e + e \mid e - e \mid e \wedge e \mid e \vee e$ $\quad e e \mid \text{if } e \text{ then } e \text{ else } e$ $\quad \text{zero?}(e)$ $\quad \text{inj.l}(e) \mid \text{inj.r}(e)$ $\quad \text{new}(e) \mid \text{get}(e) \mid \text{set}(e, e)$	$v ::= 0 \mid -1 \mid 1 \mid \dots \mid \text{tt} \mid \text{ff} \mid x$ $\quad \lambda x. e \mid \text{loc}$ $E ::= \square$ $\quad E + e \mid v + E \mid E - e \mid v - E$ $\quad E \wedge e \mid v \wedge E \mid E \vee e \mid v \vee E$ $\quad E e \mid v E \mid \text{if } E \text{ then } e \text{ else } e \mid \text{zero?}(E)$ $\quad \text{inj.l}(E) \mid \text{inj.r}(E)$ $\quad \text{new}(E) \mid \text{get}(E) \mid \text{set}(E, e) \mid \text{set}(v, E)$

Figure 1: Base Language

2.2 **Contracts** Language

2.2.1 Example

2.2.2 Formal

We define **Contracts**, in Figure 2, by extending the Base language with contracts. Contracts protect an expression, ensuring that the evaluated value conform to the contracts defined.

Contracts are defined for all possible types of value. Base values, including booleans and numbers, are trivial to enforce a contract on them. While functions need to check for both its argument and return result. Not only that, functions are first-class citizen in the language, and we can form higher-order functions. A contract for function may want to “see” the argument when checking the result, such is the idea of dependent contracts. Both contracts higher-order functions, and dependent contracts semantics are discussed in [4].

The language supports contracts for tuples. Their contracts will be applied individually during execution. Mutable cells contracts follow the work of [2], where a special contract type $\text{ref}/c(\kappa)$ is used to protect the cell’s value.

$$\begin{array}{lcl}
 \boxed{\text{Contracts}} & \text{extends Base} & \\
 \tau ::= & \dots & \kappa ::= \text{flat}(e) \mid \kappa \rightarrow \kappa \mid \kappa \xrightarrow{d} (\lambda x : \tau. \kappa) \\
 & \mid \text{con}(\tau) & \mid \langle \kappa, \kappa \rangle \mid \text{ref}/c(\kappa) \\
 \\
 e ::= & \dots & v ::= \dots \\
 & \mid \text{mon}_j^{k,l}(\kappa, e) & \mid \text{guard}_j^{k,l}(\kappa, v) \\
 & \mid \text{check}_j^k(e, v) & \\
 & \mid \text{blame}_p^l & \\
 \\
 E ::= & \dots & \\
 & \mid \text{mon}_j^{k,l}(\kappa, E) & \\
 & \mid \text{check}_j^k(E, v) &
 \end{array}$$

Figure 2: **Contracts** language

When a contract violation occurs, a blame is thrown with the party, caller or callee. The semantics for correct blaming has been studied under [2] and [3]. Therefore, we use *indy* semantics for dependent contracts.

The type system is straightforward, readers interested in the type rules for **Contracts** can consult the Appendix. We provide the semantics for the language in Figure 3.

$\text{check}_j^k(\text{tt}, v)$	$\longrightarrow$	v	Check-True
$\text{check}_j^k(\text{ff}, v)$	$\longrightarrow$	blame_j^k	Check-False
$\text{guard}_j^{k,l}(v, \text{flat}(e))$	$\longrightarrow$	$\text{check}_j^k(e\ v, v)$	Mon-Flat
$\text{guard}_j^{k,l}(v_1, \kappa_1 \rightarrow \kappa_2)\ v_2$	$\longrightarrow$	$\text{mon}_j^{k,l}(\kappa_2, v_1\ \text{mon}_j^{l,k}(\kappa_1, v_2))$	Guard-Func
$\text{guard}_j^{k,l}\left(v_1, \kappa_1 \xrightarrow{d} (\lambda x. \kappa_2)\right) v_2$	$\longrightarrow$	$\text{mon}_j^{k,l}(\kappa'_2, v_1\ \text{mon}_j^{k,l}(\kappa_1, v_2))$	Mon-Dep
where $\kappa'_2 = \{\text{mon}_j^{l,j}(\kappa_1, v_2)/x\}\kappa_2$			
$\text{mon}_j^{k,l}(\langle \kappa_1, \kappa_2 \rangle, \langle v_1, v_2 \rangle)$	$\longrightarrow$	$\langle \text{mon}_j^{k,l}(\kappa_1, v_1), \text{mon}_j^{k,l}(\kappa_2, v_2) \rangle$	Mon-Tuple
$\text{mon}_j^{k,l}(\text{ref}/c(\kappa), v)$	$\longrightarrow$	$\text{guard}_j^{k,l}(\kappa, v)$	Mon-Cell
$\text{get}(\text{guard}_j^{k,l}(\kappa, v))$	$\longrightarrow$	$\text{mon}_j^{k,l}(\kappa, \text{get}(v))$	Mon-Get
$\text{set}(\text{guard}_j^{k,l}(\kappa, v_1), v_2)$	$\longrightarrow$	$\text{mon}_j^{k,l}(\text{ref}/c(\kappa), \text{set}(v_1, \text{mon}_j^{l,k}(\kappa, v_2)))$	Mon-Set
$\frac{E \neq \square}{E[\text{blame}_j^k] \Rightarrow \text{blame}_j^k} \text{Blame}$			

Figure 3: Reduction Rules for **Contracts**

2.3 **Effects** Language

2.3.1 Example

2.3.2 Formal

We continue to extend the Base language with effects and effect handlers in Listing 1. This is done by typing the effects separately from the base types. This practice has been introduced to formalize the Koka¹ language. The approach we take here resembles the System F^ε language introduced in [14]. We added polymorphism through type application, and kinds to Base language. Two special kinds are added to represent effects (**eff**) and effect labels (**lab**). A label $l \in \mathbf{lab}$ is a group of operations. And an effect $e \in \mathbf{eff}$ is composed by chaining **lab** together with other effects. An empty effect $\langle \rangle$ defines no labels,

¹<https://koka-lang.github.io/>

therefore no effects can be performed. In otherwords, **eff** defines the set of effects that can be performed. A function can now produce some effects $\varepsilon \in \mathbf{eff}$, and is represented in its type $\tau_1 \rightarrow^\varepsilon \tau_2$. Original type for function $\tau_1 \rightarrow \tau_2$ is understood as pure, without effects, $\tau_1 \rightarrow^\langle \rangle \tau_2$.

The language is then extended with handlers, handler installation (**handle** h e), and calling/performing an effect (**perform** op τ). Handlers are a set of operations distinguishable by the operation's name. A handler denotes a single effect usually by its label $l \in \mathbf{lab}$. Performing an effect is analogous function application, therefore **perform** op τ is treated as a value, where application for it has a distinct reduction rule. The evaluation context is thus extended to support handler installation.

Effects extends Base

$\tau ::= \dots$	$v ::= \dots$
α^k $c^k \tau \dots$ $\tau \rightarrow^\varepsilon \tau$ $\forall \alpha^k. \tau$	$\Lambda \alpha^k. v$
	handler h
$k ::= *$	perform op τ
$k \rightarrow k$	
eff	$E ::= \dots$
lab	$E[\tau]$
	handle h E
$e ::= \dots$	$F ::= \square$
$e[\tau]$	$F + e$ $v + F$ $F \wedge e$ $v \wedge F$ $F \vee e$ $v \vee$
handle h e	F
blame _{p}	$F e$ $v F$
$h ::= \{\text{op}_1 \rightarrow f_1, \dots, \text{op}_n \rightarrow f_n\}$	if F then e else e
	$F[\tau]$

Listing 1: **Effects** language

Typing for an effect language requires some consideration. Following the works in [14], we define the type judgement relationship $\Delta \vdash e : \tau \mid \varepsilon$ between a type context Δ , an expression e and assert it against type τ with some possible effects ε . Variables and base

values do not have effects, they can take any effects. For this reason, we define $\Delta \vdash_{\text{val}} v : \tau$. Handlers are typed using $\Delta \vdash_{\text{ops}} h : \tau \mid l \mid \varepsilon$, that it handles a *single* effect l , other effects in ε remains unhandled, and all operations returns τ . All type rules unique to **Effects** is presented in Figure 4.

$$\begin{array}{c} \vdots \\ \text{--- SomeRule} \\ \vdots \end{array}$$

Figure 4: Type Rules for **Effects**

Lastly the semantic model of **Effects** is presented in Listing 2. A handler can be installed to a function as a shorthand, where it applies the function with a unit argument, which to be ignored. The reduction for handling an effect calling is encoded as *deep handler*. Supporting the shallow should be straightforward by not installing the handler again in the continuation k . There is little advantage to support shallow handlers, therefore, only deep handlers are used. While looking for the handler, we make sure we get the innermost handler that can handle the operation.

$$\begin{array}{lll}
(\Lambda \alpha^k . v)[\tau] & \Longrightarrow & v[\alpha := \tau] \quad \text{Step-E-TApp} \\
\text{handler } h \ v & \Longrightarrow & \text{handle } h \ (v \ ()) \quad \text{Step-E-Handler} \\
\text{handle } h \ v & \Longrightarrow & v \quad \text{Step-E-Return} \\
\\
\text{op} \notin \text{bop}(E) \wedge (\text{op} \rightarrow f) \in h \quad \text{op} : \forall \alpha . \tau_1 \rightarrow \tau_2 \in \Sigma(l) \\
\frac{k = \lambda x : \tau_2 [\alpha := \tau] . \text{handle } h \ E[x]}{\text{handle } h \ E[\text{perform op } \tau \ v] \Longrightarrow f[\tau] \ v \ k} \quad \text{Step-E-Perform} \\
\\
\frac{E \neq \square}{E[\text{blame}_p] \Longrightarrow \text{blame}_p} \quad \text{Effect-Error} \\
\\
\textbf{Helper functions} \\
\begin{array}{lll}
\text{bop}(\square) & = & \emptyset \\
\text{bop}(E \ e) & = & \text{bop}(E) \\
\vdots & & \vdots \\
\text{bop}(\text{handle } h \ E) & = & \text{bop}(E) \cup \{\text{op} \mid (\text{op} \rightarrow f) \in h\}
\end{array}
\end{array}$$

Listing 2: Reduction Rules for **Effects**

CHAPTER 3

Formal Compiler

In this section, we discuss a compiler from **Contracts** to **Effects**. It is straightforward that many expressions are inherited from the Base language, and the compiler keeps these same constructions. What is left are contract-related expressions. To reiterate our idea that a contract is a side effect, we define such effect as $\mathcal{C}$. This effect ideally receives a tuple of $\langle l, \langle e, v \rangle \rangle$ where l is the blame label, e is the contract predicate, and v is the protected value. Then, a flat contract [Compile-Flat] is simply an invocation of such effect with its correct inputs.

Functional contracts are more complex. Because the effect can only be invoked on a flat contract, the compiler recursively expands the expression, pads the expression into a function when it is expecting a function. During the execution, new variables are generated to pad the expression into a function. The process is also applicable to dependent contracts. However, the dependent contract needs to update the post contract with a contract for argument. This is trivial but requires a duplication of compilation with a different label.

After all flat contracts are converted into performing an effect, we need a handler. A simple handler can be installed like below:

$$h = \{\mathcal{C} \rightarrow \lambda \langle l, \langle e, v \rangle \rangle. \lambda k. \text{ if } e \ v \text{ then } k \ v \text{ else } \text{blame}_l\}$$

where $\lambda \langle l, \langle e, v \rangle \rangle$ deconstructs the value into tuple pattern

This handler performs the check for the value v , with the predicate/contract e . The continuation k is invoked with the value if the check passes, and blame the label l if the contract is violated. At first glance, this handler is correct. In fact, the handler should produce correct semantic, as long as no dependent contracts are used. When dependent

contracts are used, e is now populated with monitored arguments, which after compilation should include **perform** $\mathcal{C}$ τ arg. And since the handler is moved into the continuation k , there is no handler at $e v$.

TODO: add example here

To resolve this issue, we define a recursive function `wrap` that will reinstall the handler at $e v$ making sure all effects during contract checking is handled. All compilation rules are defined in Figure 5. With `wrap` defined, we can make some theorems about the compiler.

Theorem 3.1. If $\cdot \vdash e_1 : \tau \rightsquigarrow e_2$ then $\cdot \vdash e_2 : \tau \mid \langle \mathcal{C} \rangle$.

$$\begin{array}{c}
 \frac{}{\Gamma \vdash \text{mon}_j^{k,l}(\text{flat}(e), v) : \tau \rightsquigarrow \text{perform } \mathcal{C} \tau \langle k, \langle e, v \rangle \rangle} \text{Compile-Flat} \\
 \\
 \frac{\begin{array}{c} x \notin \Gamma \quad \Gamma \vdash \text{mon}_j^{l,k}(\kappa_1, x) : \tau_1 \rightsquigarrow x_1 \\ \Gamma, x : \tau_1 \vdash \text{mon}_j^{k,l}(\kappa_2, (\lambda x : \tau_1. e) x_1) : \tau_2 \rightsquigarrow e_1 \end{array}}{\Gamma \vdash \text{mon}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, e) : \tau_1 \rightarrow \tau_2 \rightsquigarrow e_1} \text{Compile-Func} \\
 \\
 \frac{\begin{array}{c} x \notin \Gamma \quad \Gamma \vdash \text{mon}_j^{l,k}(\kappa_1, x) : \tau_1 \rightsquigarrow x_1 \quad \Gamma \vdash \text{mon}_j^{l,j}(\kappa_1, x) : \tau_1 \rightsquigarrow x_2 \\ \kappa_3 = \{x_2/x\}\kappa_2 \quad \Gamma, x : \tau_1 \vdash \text{mon}_j^{k,l}(\kappa_3, (\lambda x : \tau_1. e) x_1) : \tau_2 \rightsquigarrow e_1 \end{array}}{\Gamma \vdash \text{mon}_j^{k,l}\left(\kappa_1 \xrightarrow{d} \lambda x. \kappa_2, e\right) : \tau_1 \rightarrow \tau_2 \rightsquigarrow e_1} \text{Compile-Dep} \\
 \\
 \hline
 \text{wrap} = \mu w : \forall \alpha. ((() \rightarrow^{\langle \mathcal{C} \rangle} \alpha) \rightarrow \alpha). \Lambda \alpha. \lambda f : (() \rightarrow^{\langle \mathcal{C} \rangle} \alpha). \\
 \quad \text{handle } \{\mathcal{C} \rightarrow \lambda \langle l, \langle e, v \rangle \rangle. \lambda k. \text{ if } w[\text{bool}] (\lambda(). e v) \text{ then } k v \text{ else blame}_l\} f \\
 \quad \text{where } \lambda \langle l, \langle e, v \rangle \rangle \text{ deconstructs the argument as tuple pattern}
 \end{array}$$

Figure 5: Compiler Rules

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CHAPTER A

Contract Erasure

CHAPTER B

Effects Safety

CHAPTER C

Compiler Soundness

Theorem 3.1. If $\cdot \vdash e_1 : \tau \rightsquigarrow e_2$ then $\cdot \vdash e_2 : \tau \mid \langle \mathcal{C} \rangle$.

Proof.

□

Theorem 3.1. If $\cdot \vdash e_1 : \tau \rightsquigarrow e_2$ then $\cdot \vdash \text{wrap}[\tau] e : \tau \mid \langle \rangle$.

Proof.

□